A Implementation details

A.1 Part of Speech annotations

Note that none of the datasets used here come annotated with word class information. We adopt the Universal Dependencies tagset, using Stanza (Qi et al., 2020, v.1.8.2) to tag words with their Universal Dependencies parts of speech. We remove single orthographic words that Stanza assigns multiple parts of speech, like English "don't" or German "zum" from our analysis, since it is unclear to which part of speech they should be assigned. Stanza does not cover Thai, Maori, Tagalog, Swahili, or Bengali for part of speech tagging, so they are excluded.

A.2 Word-level PMI Estimates

Because the tokenizer of the present model does not cross orthographic word boundaries, we are able to sum the log probabilities of their constituent subword tokens to obtain word-level rather than token-level log probability estimates. Ordinarily, some languages do not indicate word boundaries in their orthography, such as Japanese; however, the pretraining data and evaluation datasets (Crossmodal-3600 and COCO-35L) are word-tokenized, so this information is readily available. Further, because our language model uses sub-word tokenization with trailing whitespaces, we adopt the correction proposed by Oh and Schuler (2024); Pimentel and Meister (2024). Specifically, let $\mathbf{s}_{w_t}$ be the decomposition of word w_t into a sequence of subwords, and $\mathbf{s}_{\mathbf{w}_{<t}}$ be the decomposition of context $\mathbf{w}_{<t}$ into a sequence of subwords. Given $\mathcal{S}_{\text{bow}}$, the subset of the tokenizer vocabulary that contains subwords that are beginning-of-word (e.g., with a trailing whitespace):

$$p(w_t \mid \mathbf{w}_{<t}) = p(\mathbf{s}_{w_t} \mid \mathbf{s}_{\mathbf{w}_{<t}}) \cdot \frac{\sum_{s \in \mathcal{S}_{\text{bow}}} p(s \mid \mathbf{s}_{\mathbf{w}_{<t}} \odot \mathbf{s}_{w_t})}{\sum_{s \in \mathcal{S}_{\text{bow}}} p(s \mid \mathbf{s}_{\mathbf{w}_{<t}})} \tag{5}$$

where $\odot$ stands for concatenation.

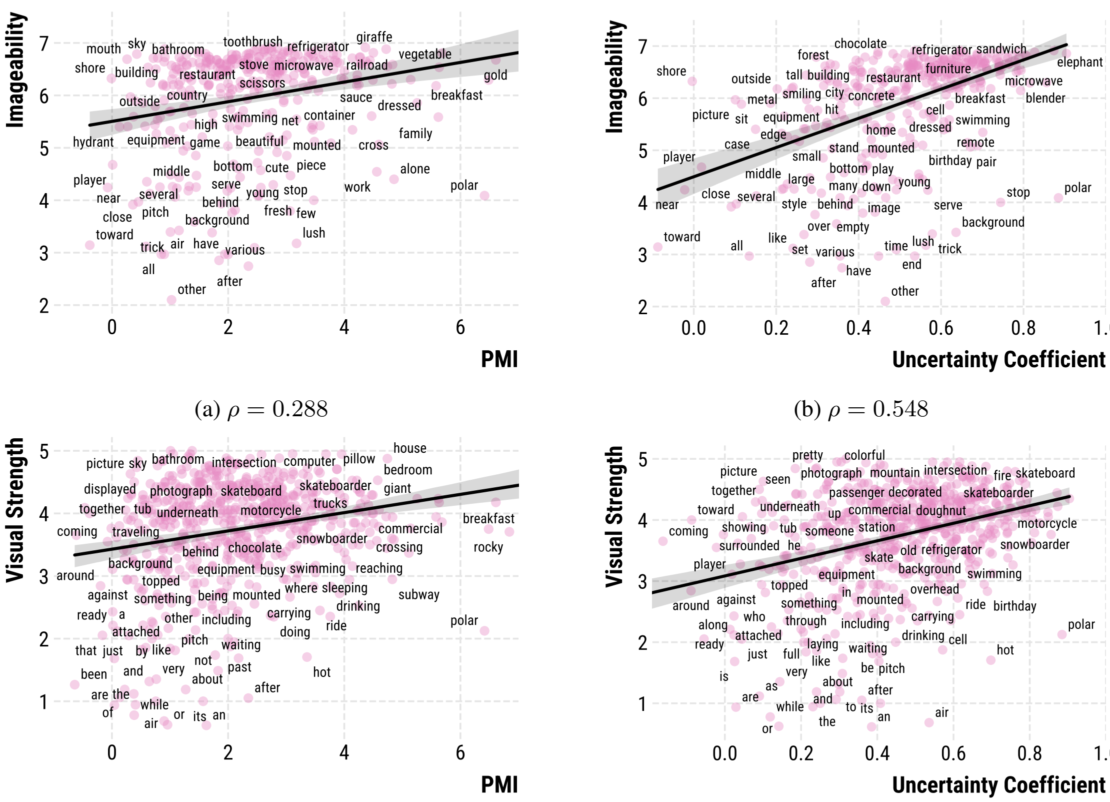
A.3 Training details

For training our language model, we did a grid search over learning rates and whether or not to use weight decay. We use a learning rate of 2×10^{-5} and weight decay of 1×10^{-6} with the Adam optimizer. To train the final model, we train on a single A100 with a batch size of 4 for 430,000 steps on COCO-35L ($\approx$ 50 hours of training, approximately 3 epochs). Our model achieves much lower perplexity on our evaluation datasets than Gemma-2B, suggesting successful domain adaptation.

B F-statistics

C Correlation plots for other psycholinguistic norms

Figure 5 shows the relationship between our measure and concreteness, as well as the uncertainty coefficient, which normalizes our measure by the language model surprisal. While concreteness is most strongly associated with our measure/its normalized variant, for completeness we show the relationships between our measure and the other psycholinguistic norms (imageability and strength of visual experience) we investigate here.



(a) $\rho = 0.288$ (b) $\rho = 0.548$

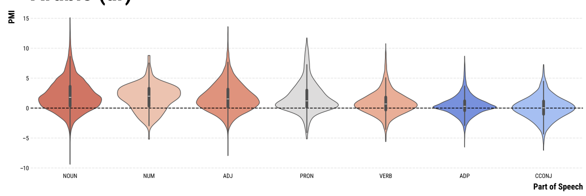
(c) $\rho = 0.212$ (d) $\rho = 0.320$

Figure 5: Correlation between English psycholinguistic norms and type-level groundedness (left) or uncertainty coefficent (right): i.e., the average ratio between LM surprisal and captioning model surprisal. Type-level measures were computed by averaging scores across the COCO-dev dataset for types which occur at least 30 times.

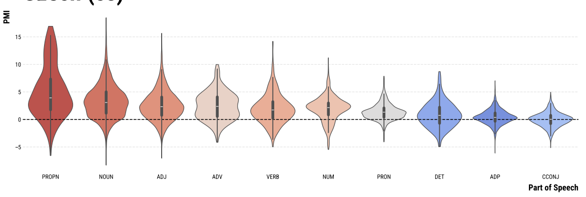
D Groundedness distribution for Crossmodal-3600

Results are ordered by descending mutual information estimate within the dataset (average groundedness/PMI). Hue indicates the average cross-linguistic ranking of a part of speech.

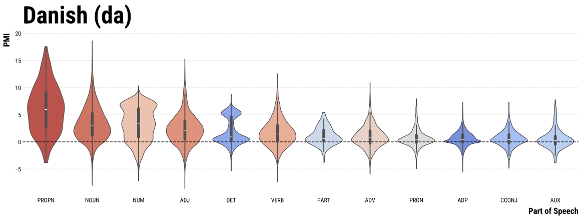
Arabic (ar)



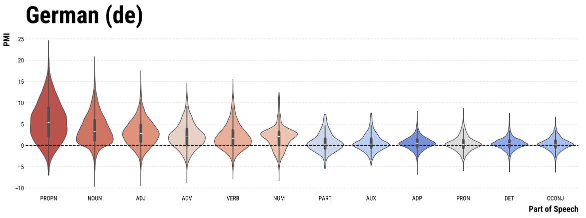
Czech (cs)



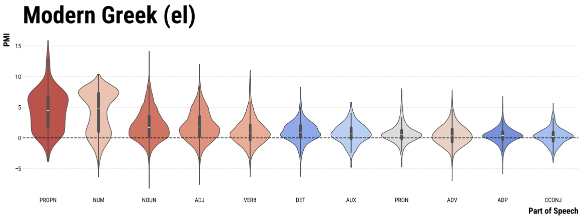
Danish (da)



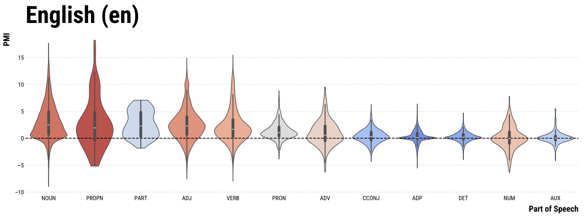
German (de)



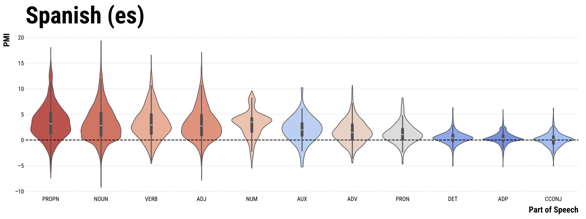
Modern Greek (el)



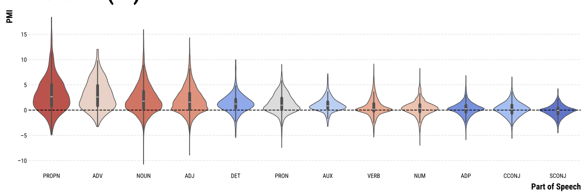
English (en)



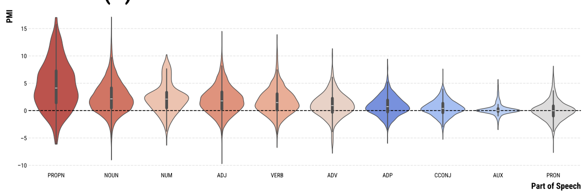
Spanish (es)



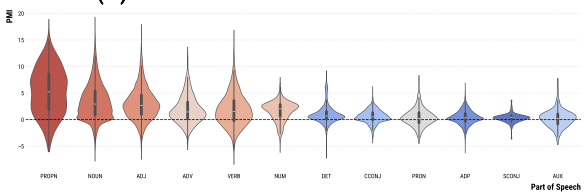
Persian (fa)



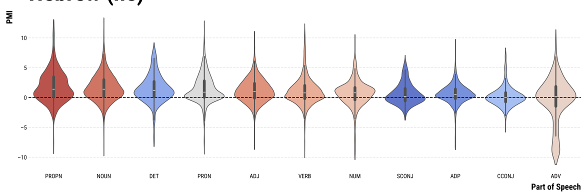
Finnish (fi)



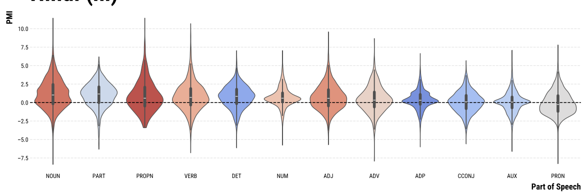
French (fr)



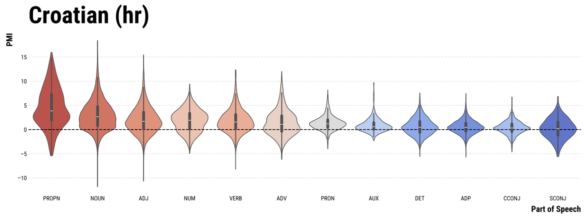
Hebrew (he)



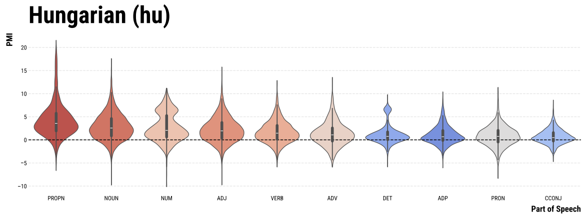
Hindi (hi)



Croatian (hr)



Hungarian (hu)



Indonesian (id)